

§4-6

Where is wrong with the following calculation?

$$\begin{aligned}
 \lim_{x \rightarrow 0} \frac{\tan x - x}{x^3} &\stackrel{(1)}{=} \lim_{x \rightarrow 0} \frac{\frac{\sin x}{\cos x} - x}{x^3} \stackrel{(2)}{=} \lim_{x \rightarrow 0} \frac{\sin x - x \cos x}{x^3 \cos x} \\
 &\stackrel{(3)}{=} \lim_{x \rightarrow 0} \frac{x \left(\frac{\sin x}{x} - \cos x \right)}{x^3 \cos x} \stackrel{(4)}{=} \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2 \cos x} \\
 &\stackrel{(5)}{=} \lim_{x \rightarrow 0} \frac{(1 - \cos x)(1 + \cos x)}{x^2 \cos x (1 + \cos x)} \stackrel{(6)}{=} \lim_{x \rightarrow 0} \frac{\sin^2 x}{x^2 \cos x (1 + \cos x)} \\
 &\stackrel{(7)}{=} \lim_{x \rightarrow 0} \frac{1}{\cos x (1 + \cos x)} \\
 &\stackrel{(8)}{=} \frac{1}{2}
 \end{aligned}$$

從 (3) 到 (4) 出了問題。

$$\text{令 } \frac{x}{x^3 \cos x} = f(x), \quad \frac{\sin x}{x} = g(x), \quad \frac{x \cos x}{x^3 \cos x} = \frac{1}{x^2} = h(x)$$

$$\text{則 (3) 可寫成 } \lim_{x \rightarrow 0} (f(x)g(x) - h(x)).$$

仔細看看從 (3) 如何變成 (4)。

$$\begin{aligned}
 \lim_{x \rightarrow 0} (f(x)g(x) - h(x)) &\stackrel{(i)}{=} \lim_{x \rightarrow 0} (f(x)g(x)) - \lim_{x \rightarrow 0} h(x) \\
 &\stackrel{(ii)}{=} \lim_{x \rightarrow 0} f(x) \lim_{x \rightarrow 0} g(x) - \lim_{x \rightarrow 0} h(x) \\
 &\stackrel{(iii)}{=} \lim_{x \rightarrow 0} f(x) - \lim_{x \rightarrow 0} h(x) \\
 &\stackrel{(iv)}{=} \lim_{x \rightarrow 0} (f(x) - h(x)) \\
 &\stackrel{(v)}{=} \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2 \cos x}
 \end{aligned}$$

請注意 $\lim_{x \rightarrow 0} f(x)$, $\lim_{x \rightarrow 0} h(x)$ 皆不存在，因此上述運算皆不合法。

正確的算法可由 L'Hospital's Rule 得

$$\begin{aligned}
 \lim_{x \rightarrow 0} \frac{\tan x - x}{x^3} &= \lim_{x \rightarrow 0} \frac{\sec^2 x - 1}{3x^2} \\
 &= \lim_{x \rightarrow 0} \frac{2 \sec^2 x \tan x}{6x} \\
 &= \lim_{x \rightarrow 0} \frac{\sec^2 x}{3} \left(\frac{\sin x}{x} \right) \\
 &= \lim_{x \rightarrow 0} \frac{\sec^2 x}{3} = \frac{1}{3} .
 \end{aligned}$$